DBMLLA: Double-Branch Mamba-Like Linear Attention Network for Hyperspectral Image Classification

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***Abstract*—Convolutional neural networks (CNNs) and transformers have made remarkable achievements in hyperspectral image classification (HSIC). Unfortunately, CNN-based methods struggle to capture the contextual dependencies between pixels in HSIs, while transformer-based methods suffer from quadratic computational complexity. Recently, the Mamba model has shown great potential as it can describe long-range dependencies between HSI pixels with linear computational complexity. Yet, Mamba still faces significant challenges in terms of global modeling. Inspired by the Mamba model framework and transformers, a novel double-branch Mamba-like linear attention (DBMLLA) network is proposed for HSIC, achieving efficient global dependency modeling. Specifically, the proposed DBMLLA combines an embedding module, a spatial–spectral Mamba-like linear attention (SS-MLLA) module, and a fusion module. In the embedding module, an absolute position embedding module is introduced for better extraction of global features. In the SS-MLLA module, we design the spatial MLLA (SpaMLLA) block and the spectral MLLA (SpeMLLA) block to exploit the spatial and spectral information of the HSI. In addition, SS-MLLA is improved using depthwise separable convolution (DSC) to enhance the model's ability to extract deeper local feature information. Through experiments conducted on four public hyperspectral datasets, it is demonstrated that the proposed model consistently outperforms state-of-the-art approaches.**

***Index Terms*—Absolute positional encoding, depthwise separable convolution (DSC), hyperspectral image classification (HSIC), Mamba, Mamba-like linear attention (MLLA) module.**

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I. Introduction

HYPERSPECTRAL imaging (HSI) captures information about the light reflected by objects across multiple spectral bands [1]. Unlike traditional RGB imagery, HSI collects data over a much wider range of the spectrum. Each pixel in an HSI contains not only spatial information but also the spectral characteristics of the corresponding point, providing a richer level of detail [2]. Therefore, HSI has found widespread applications in geological exploration [3], environmental monitoring [4], cancer diagnosis [5], food safety [6], and more. As a fundamental task in the aforementioned applications, hyperspectral image classification (HSIC) aims to categorize each pixel in the image by predicting its class label [7].

Traditional HSIC methods are largely dependent on spectral information and statistical principles, including maximum likelihood classification (MLC) [8], minimum distance classification (MVC) [9], linear discriminant analysis (LDA) [10], support vector machine (SVM) [11], principal component analysis (PCA) [12], etc. To enhance the discriminative ability of the feature in the HSI, Kang et al. [13] introduced a novel spectral representation method called 2-D spectral representation. These methods primarily use the characteristics of the spectral band. To fully leverage the spatial data in HSI, various methods have been proposed, such as superpixel segmentation [14], multiple kernel learning (MKL) [15], and extended morphological profile (EMP) [16]. However, the descriptive power of these methods is limited because they rely heavily on the setting of prior parameters.

During the past few years, the advancement of deep learning technology has led to significant progress in HSIC [17], [18]. Representative models mainly include convolutional neural networks (CNNs) [19], deep autoencoders (AEs) [20], recurrent neural networks (RNNs) [21], deep residual networks (ResNets) [22], and graph convolutional networks (GNNs) [23]. Since CNNs can extract multilevel features from shallow to deep layers in HSI, a variety of CNN-based methods have been designed for spatial–spectral feature learning, such as 1-D CNN [24] and 2D-3D CNN [25]. Wang et al. [26] introduced dense connections, allowing deep feature extraction with a minimal number of convolutional layers, thus reducing computational complexity. To address the gradient vanishing phenomenon during training and improve model generaliza-

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tion, Paoletti et al. [27] introduced ResNets. Alkhatib et al. [28] presented a three-branch structure that uses convolutional kernels of different scales to extract features. To reduce the number of learnable parameters, Lu et al. [29] proposed a separable deep GNN that extracts effective structural information from HSI to achieve good classification performance, without increasing the number of trainable parameters. Moreover, Duan et al. [30] proposed an intrinsic decomposition model for shadow removal, which can obtain high-quality HSI. Nevertheless, CNNs are constrained by the local receptive field of their convolutional kernels and are unable to model long-range dependencies, which makes extracting global features challenging.

Benefiting from the attention mechanism's powerful global modeling capabilities, transformers have found widespread application in HSIC [31]. Hong et al. [32] introduced SpectralFormer, which was the first to apply a pure transformer (without any preprocessing) to HSIC tasks. However, this method ignores spatial information. Subsequently, Xu et al. [33] introduced the cross spatial–spectral dense transformer, which uses a dense learning strategy to extract spatial and spectral features while using a cross-attention module for feature fusion. Sun et al. [34] introduced a dual-branch transformer that incorporates a Gaussian weighting method to effectively extract both deep spectral and spatial features simultaneously. Xu et al. [35] proposed a cross-attention fusion network, which uses a dynamic CNN and a Gaussian transformer to encode local features and long-range dependencies of pixels, respectively, and uses a cross-attention fusion module as a bridge to effectively fuse the features. Shi et al. [36] presented the multiscale transformer to capture spatial and spectral information at multiple scales. Li et al. [37] incorporated spectral prior information into the attention mechanism to enhance global feature extraction. Although transformers have shown impressive capabilities in HSIC, the inherent self-attention mechanism of transformers has quadratic computational complexity [38], which makes them inefficient when processing long sequence data, posing both memory and speed challenges.

Recently, Mamba, built on state-space models (SSMs) [39], has demonstrated significant potential in HSIC by modeling contextual dependencies while maintaining linear computational complexity [40]. Gu and Dao [41] first proposed the Mamba and described its superior performance over transformers. Li et al. [42] pioneered the application of Mamba in HSIC, introducing the first image-level HSIC method based on SSM, which achieved promising classification results. Because of the limitations of the Mamba algorithm in causal sequence modeling and its limitations in handling high-dimensional scenes, He et al. [43] developed a 3-D selective scanning mechanism that separately captures spatial and spectral features from the perspective of sequence modeling, thus achieving feature extraction for 3-D hyperspectral data. Yao et al. [44] proposed SpectralMamba, which was the first work to customize deep SSM specifically for hyperspectral data and its analysis. Huang et al. [45] devised a token generation mechanism to facilitate the efficient integration of spatial–spectral data. To overcome information redundancy, He et al. [46] introduced IGroupSS-Mamba, which performs multidirectional global feature extraction in a grouped manner. Although Mamba has advantages in computational efficiency, transformers generally perform better in tasks that require global contextual information [47].

To overcome the above problems, this research presents a novel double-branch Mamba-like linear attention (DBMLLA) network, an HSIC framework that is based on linear attention, with a structure similar to the Mamba block, for the first time. Using the linear attention transformer, the model captures both fine-grained local details and broad global patterns of the image while maintaining linear computational complexity. DBMLLA consists of an embedding module, a spatial–spectral Mamba-like linear attention (SS-MLLA) module, a fusion module, and a classification head. Specifically, the entire image is first fed into the position embedding module, where absolute position embedding is introduced to enhance the correlation between image pixels. Then, the SS-MLLA module extracts spatial and spectral information. The SS-MLLA module uses a dual-branch framework, where one branch processes spatial information and the other handles spectral features. The spatial branch is composed of multiple stacked spatial MLLA (SpaMLLA) blocks for extracting complex spatial features in HSI, while the spectral branch is made up of multiple stacked spectral MLLA (SpeMLLA) blocks to extract spectral information. The fusion module is then exploited to integrate the spectral and spatial feature information. In addition, the SS-MLLA module is improved using depthwise separable convolution (DSC) to enhance the model's ability to extract deeper features. Finally, the fused features are fed into the classification head to generate the final classification results. Compared with the existing methods, the proposed DBMLLA demonstrates exceptional capability in the task of extracting global contextual features.

The primary contributions are concluded as follows.

1) In this article, a novel HSIC network based on a linear attention transformer with a structure similar to the Mamba block is proposed for the first time, which can efficiently achieve global contextual dependency modeling of spatial and spectral features in HSIs.
2) SpaMLLA and SpeMLLA are designed to capture complex spatial and spectral characteristics, respectively. Among them, SpaMLLA is used for modeling long-range interactions of HSI with linear complexity. Unlike SpaMLLA, SpeMLLA groups spectral features to capture correlations between different spectral groups.
3) In the SS-MLLA module, DSC is used for the first time in the MLLA module to enhance the ability of the model to extract deeper level features.
4) An absolute position embedding module is used to provide a structured representation of the input sequence, enabling the model to better capture long-range pixel dependencies.

The remaining sections of this article are outlined as follows. A brief overview of related algorithms is given in

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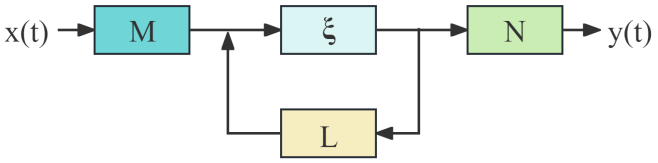


Fig. 1. SSMs' structure diagram.

Section II. The proposed method is discussed in Section III. Section IV discusses the experimental results obtained from four HSI datasets. Section V concludes this article with a summary and potential future directions.

II. Related Works

This section presents a concise overview of the algorithms associated with Mamba and absolute position encoding. Regarding Mamba, we start with an introduction to SSMs, followed by the introduction of structured SSMs through discretization, then add a selectivity mechanism to form S6, and finally form the Mamba model based on S6.

A. State-Space Models

The SSMs were first introduced by Kalman, and particularly the introduction of the Kalman filter made the SSMs widely applied [48]. SSMs can recursively process sequential data, capturing the dynamic features of the structure, and thus modeling the spatial and spectral features in HSI [39]. Numerous studies have shown that SSMs can effectively replace CNNs and transformers. A detailed depiction of the structure of SSMs can be found in Fig. 1.

The system takes $x(t) \in \mathbb{R}$ as input and maps the function or sequence to the output $y(t) \in \mathbb{R}$ via an hidden state $\xi(t) \in \mathbb{R}^N$. The process can be described by the following equation:

$$\begin{aligned} \xi'(t) &= \mathbf{L}\xi(t) + \mathbf{M}x(t) \\ y(t) &= \mathbf{N}\xi(t) \end{aligned} \tag{1}$$

where these two equations represent the state evolution and the observation process, respectively. In the equations, $x(t)$ denotes the state vector at time t, containing the system's internal information. $\mathbf{L}$ is the state transition matrix, which defines the dynamic evolution relationship of the system state from time t to time $t+1$. $\mathbf{M}$ is the input matrix, which describes how the input $x(t)$ affects the system's state. $\mathbf{N}$ is the observation matrix, which maps the system's state $x(t)$ to the observation space, describing the relationship between the state and the observation. $y(t)$ is the observation vector, representing the measurable output. Since we typically do not have direct access to the system's internal state $x(t)$, we need the observation equation to obtain the output.

B. Structured SSMs

Equation (1) describes a linear continuous system. However, in HSIC, the data typically consist of discrete sequences, so the basic SSM faces significant challenges when applied to HSIC. By adding the discretization parameter Δ to the SSM [49], the linear parameters $\mathbf{L}$ and $\mathbf{M}$ are transformed into discrete parameters $\overline{\mathbf{L}}$ and $\overline{\mathbf{M}}$, forming structured SSMs (S4). The discrete system can then be represented by the following equations:

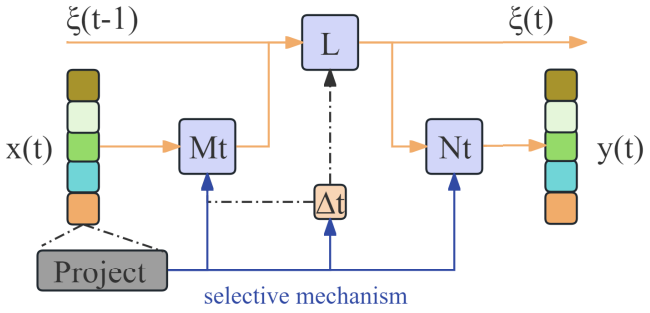


Fig. 2. Selective structured SSMs (S6) diagram.

$$\begin{aligned} \overline{\mathbf{L}} &= \exp(\Delta\mathbf{L}) \\ \overline{\mathbf{M}} &= (\Delta\mathbf{L})^{-1}\left(\exp(\Delta\mathbf{L}) - \mathbf{I}\right) \cdot \Delta\mathbf{M} \approx \Delta\mathbf{M} \\ \xi'_t &= \overline{\mathbf{L}}\xi_{t-1} + \overline{\mathbf{M}}x_t \\ y_t &= \mathbf{N}\xi_t. \end{aligned} \tag{2}$$

After the discretization of $\overline{\mathbf{L}}$ and $\overline{\mathbf{M}}$, the process can be accelerated using convolution operations, which improves scalability and efficiency. The final output can be represented by the following equations:

$$\begin{aligned} \overline{\mathbf{K}} &= \left(\mathbf{N}\overline{\mathbf{M}}, \mathbf{N}\overline{\mathbf{L}}\overline{\mathbf{M}}, \ldots, \mathbf{N}\overline{\mathbf{L}}^{\mathbf{S}-1}\overline{\mathbf{M}}\right) \\ \mathbf{y} &= \mathbf{x} * \overline{\mathbf{K}} \end{aligned} \tag{3}$$

where $\mathbf{S}$ represents the size of the input sequence x, and $\overline{\mathbf{K}} \in \mathbb{R}^S$ represents the structured convolution kernel.

C. Selective Structured SSMs

To accurately capture the contextual dependencies in the sequence, S4 introduces a selective mechanism to form a selective structured SSM (S6). The selective mechanism parameterizes the input of the SSM, allowing the model to selectively process information to focus on or ignore specific inputs [50]. The structure diagram of S6 is shown in Fig. 2.

This selective mechanism is similar to the self-attention in transformers, but can significantly improve computational efficiency because of its lower computational complexity.

D. Mamba Block

Given Mamba's capability for modeling high-resolution images, it has been widely used in visual tasks, containing HSIC.

The structure diagram of Mamba is displayed in Fig. 3(b). As revealed in the figure, the Mamba block mainly consists of the batch normalization and the SSM block. In addition, the concept of residual learning is introduced. The SSM block is primarily composed of S6, linear layers, convolutional layers, and so on. The SSM can capture the contextual dependencies between image pixels, while the convolutional layers within extract local features. Overall, the Mamba block can extract

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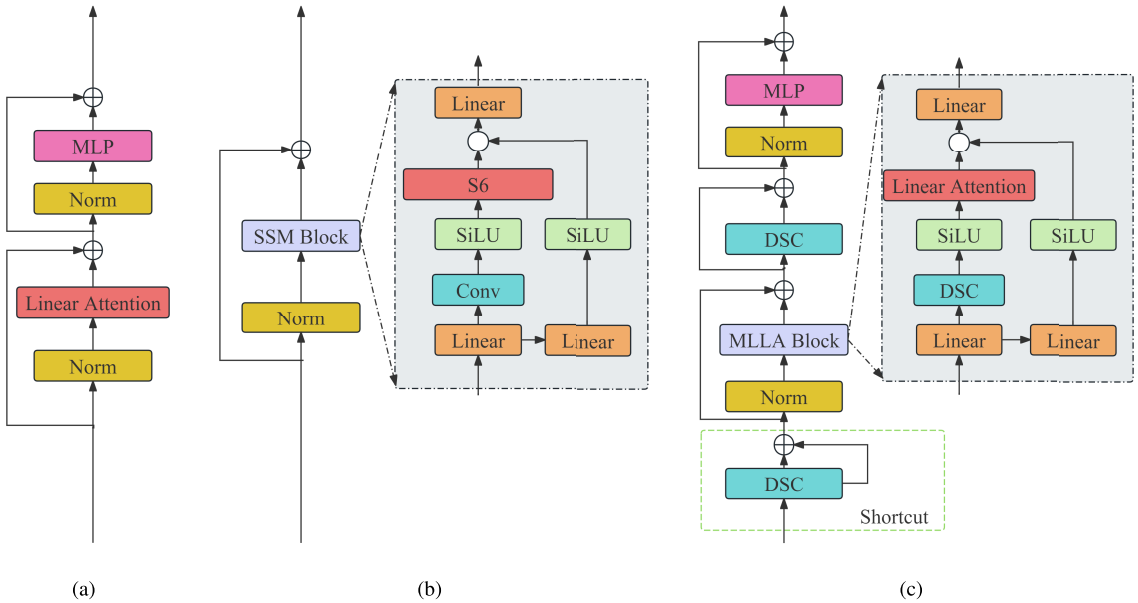


Fig. 3. (a) Linear attention transformer. (b) Mamba. (c) MLLA.

both comprehensive and specific spatial–spectral information from images, making it well-suited for applications in HSIC.

However, as a causal sequence model, Mamba generally exhibits weaker global modeling capabilities in vision tasks compared with transformers. Han et al. [51] conducted a visual analysis to reveal the similarities between Mamba and linear attention transformers and proposed a novel MLLA model that integrates the advantages of both the architectures. The MLLA model achieves enhanced global modeling capability while maintaining linear computational complexity. A detailed description of the MLLA module is provided in Section III.

E. Absolute Positional Encoding

Absolute positional encoding aims to model sequential dependencies among elements by incorporating positional embeddings into the input sequence. This mechanism enhances the model's awareness of positional correlations through elementwise embedding integration, thereby preserving contextual order information during feature learning. Each input position is assigned a vector representation that contains positional information. The position encoding is typically generated using a sine and cosine function. Specifically, given position $\boldsymbol{e}$ and dimension $\boldsymbol{d}$, the position encoding can be calculated using the formulas below

$$\begin{aligned} \mathrm{APE}_{(e,2\ d)} &= \sin\left(\frac{e}{10\,000^{\frac{2d}{D}}}\right) \\ \mathrm{APE}_{(e,2\ d+1)} &= \cos\left(\frac{e}{10\,000^{\frac{2d}{D}}}\right) \end{aligned} \tag{4}$$

where D indicates the dimension of the positional encoding. As soon as the positional encoding is generated, it is added to the input features on an element-by-element basis. This allows the model not only to process the input data but also to understand their positions in the sequence.

III. Proposed Method

Fig. 4 shows the algorithm structure diagram of the proposed DBMLLA model, which primarily consists of three stages. The first stage is the position embedding module, which enhances position embedding by incorporating absolute position encoding. The second stage uses SpaMLLA to model the global contextual dependencies of the HSI and uses SpeMLLA to capture the correlations between spectral groups, extracting spatial and spectral features simultaneously through a dual-branch structure. In the third stage, spatial and spectral information is adaptively fused using attention weights. Based on these modules, this section elaborates on the comprehensive architecture of the DBMLLA framework for HSIC.

A. MLLA Module

To model the long-range dependencies between HSI pixels, transformers are typically used for HSIC. In transformers, softmax attention serves as a key module [52]. Let $\boldsymbol{x} \in \mathbb{R}^{N\times M}$ represent N feature sequences, each with a dimensionality of M. Single-head softmax attention can be represented using the following equations:

$$\begin{aligned} \mathbf{Q} = \boldsymbol{x}\mathbf{Z}_Q,\ \mathbf{K} = \boldsymbol{x}\mathbf{Z}_K,\ \mathbf{V} = \boldsymbol{x}\mathbf{Z}_V \\ \boldsymbol{y}_m = \sum_{n=1}^{M} \frac{\exp\left(\mathbf{Q}_m\mathbf{K}_n^{\top}/\sqrt{D}\right)}{\sum_{n=1}^{M}\exp\left(\mathbf{Q}_m\mathbf{K}_n^{\top}/\sqrt{D}\right)}\mathbf{V}_n \end{aligned} \tag{5}$$

where $\mathbf{Z}_Q$, $\mathbf{Z}_K \in \mathbb{R}^{M\times D}$, and $\mathbf{Z}_V \in \mathbb{R}^{M\times M}$ are the projection matrices. $\mathbf{Q}$ and $\mathbf{K} \in \mathbb{R}^{N\times D}$ are the query matrices and key matrices, respectively, and $\mathbf{V} \in \mathbb{R}^{N\times M}$ is the value matrix. $\mathbf{Q}_m$, $\mathbf{K}_m \in \mathbb{R}^{1\times D}$, and $\mathbf{V}_n \in \mathbb{R}^{1\times M}$ represent the individual query, key, and value tokens, respectively. In Softmax attention, the similarity between each query–key pair is computed in the order of $(\mathbf{QK}^{\top})\mathbf{V}$ calculation, resulting in a computational complexity of $\mathcal{O}(N^2D + N^2M) \approx \mathcal{O}(N^2)$.

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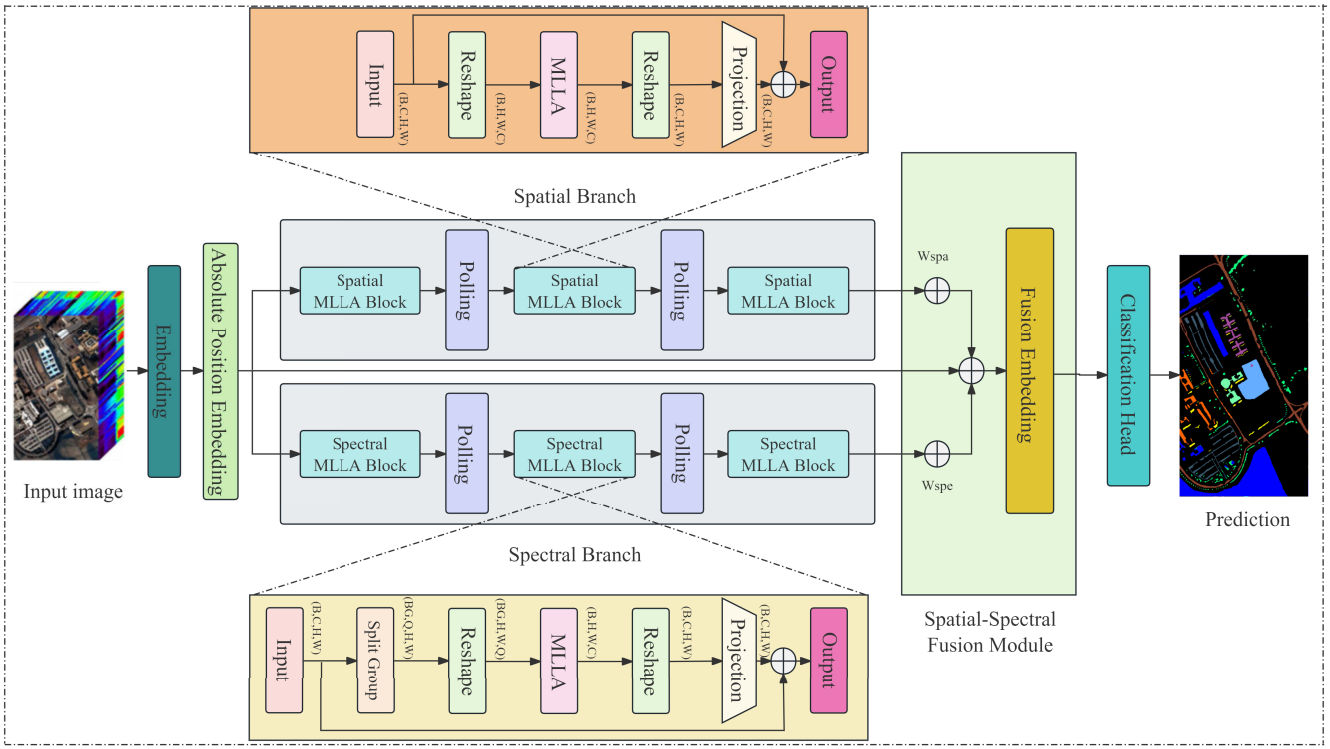


Fig. 4. Schematic of the proposed DBMLLA module. First, the HSI is fed into the embedding layer, and the embeddings are then input into SpaMLLA and SpeMLLA to capture spatial and spectral features. Finally, after feature fusion through the fusion module, the fused features are passed through the classification head to generate the final classification output.

In the context of modeling long sequences, self-attention results in quadratic computational complexity, which leads to high computational costs. To alleviate this limitation, we substitute the nonlinear softmax attention function with a linear normalization and introduce a kernel function ϕ in $\mathbf{Q}$ and $\mathbf{K}$. The formulas are detailed below

$$\mathbf{Q} = \phi\left(\boldsymbol{x}\mathbf{Z}_Q\right),\ \mathbf{K} = \phi\left(\boldsymbol{x}\mathbf{Z}_K\right),\ \mathbf{V} = \boldsymbol{x}\mathbf{Z}_V$$
$$\boldsymbol{y}_m = \sum_{n=1}^{M} \frac{\mathbf{Q}_m\mathbf{K}_n^\top}{\sum_{j=n}^{M}\mathbf{Q}_m\mathbf{K}_n^\top}\mathbf{V}_n = \frac{\mathbf{Q}_m\left(\sum_{n=1}^{M}\mathbf{K}_n^\top\mathbf{V}_n\right)}{\mathbf{Q}_m\left(\sum_{n=1}^{M}\mathbf{K}_n^\top\right)}. \tag{6}$$

This changes the computation order from $(\mathbf{QK}^\top)\mathbf{V}$ to $\mathbf{Q}(\mathbf{K}^\top\mathbf{V})$, reducing the complexity to $\mathcal{O}(NDM)$. In most cases, $D, M \ll N$, so the complexity is approximately $O(N)$, which is linear complexity.

As shown in (6), the linear attention mechanism is defined with a global receptive field, enabling each query to aggregate information from all keys and values. It does not impose constraints on the position or temporal order of the input, allowing interactions between any pieces of information. In practice, such a mechanism can also be applied in autoregressive models. We limit the acceptance domain of the mth token to the continuation tokens, token n, $n \le m$. The formulas are as follows:

$$\boldsymbol{y}_m = \frac{\mathbf{Q}_m\left(\sum_{n=1}^{m}\mathbf{K}_n^\top\mathbf{V}_n\right)}{\mathbf{Q}_m\left(\sum_{n=1}^{m}\mathbf{K}_n^\top\right)} \triangleq \frac{\mathbf{Q}_m\mathbf{U}_m}{\mathbf{Q}_m\mathbf{G}_m}$$
$$\mathbf{U}_m = \sum_{n=1}^{m}\mathbf{K}_n^\top\mathbf{V}_n$$
$$\mathbf{G}_m = \sum_{n=1}^{m}\mathbf{K}_n^\top. \tag{7}$$

From this result, the following linear attention recursive form can be derived:

$$\begin{aligned}\mathbf{U}_m &= \mathbf{U}_{m-1} + \mathbf{K}_m^\top\mathbf{V}_m\\ \mathbf{G}_m &= \mathbf{G}_{m-1} + \mathbf{K}_m^\top\\ \boldsymbol{y}_m &= \mathbf{Q}_m\mathbf{U}_m/\mathbf{Q}_m\mathbf{G}_m.\end{aligned} \tag{8}$$

For SSM formulation in (2), we make the following modifications. As Mamba represents $\mathbf{L}$ as a diagonal matrix, we have $\overline{\mathbf{L}}_i\xi_{i-1} = \tilde{\mathbf{L}}_i \odot \xi_{i-1}$, where $\tilde{\mathbf{L}}_i = \mathrm{diag}(\overline{\mathbf{L}}_i)$ denotes the vector composed of the diagonal elements of $\overline{A}_i$. According to (2), $\overline{\mathbf{M}}_i = \Delta_i\mathbf{M}_i$, it follows that $\overline{\mathbf{M}}_i x_i = \Delta_i\mathbf{M}_i x_i = \mathbf{M}_i(\Delta_i x_i) = \mathbf{M}_i(\Delta_i \odot x_i)$. Therefore, (2) can be rewritten as follows:

$$\begin{aligned}\boldsymbol{\xi}_i &= \widetilde{\mathbf{L}}_i \odot \boldsymbol{\xi}_{i-1} + \mathbf{M}_i\left(\Delta_i \odot x_i\right)\\ y_i &= \mathbf{N}_i\boldsymbol{\xi}_i.\end{aligned} \tag{9}$$

We unify (8) and (9) into a unified formulation

$$\begin{aligned}\boldsymbol{\xi}_i &= \widetilde{\mathbf{L}}_i \odot \boldsymbol{\xi}_{i-1} + \mathbf{M}_i\left(\Delta_i \odot \mathbf{x}_i\right)\\ y_i &= \mathbf{N}_i\boldsymbol{\xi}_i/1\end{aligned} \tag{10}$$
$$\begin{aligned}\mathbf{U}_m &= 1 \odot \mathbf{U}_{m-1} + \mathbf{K}_m^\top\left(1 \odot \mathbf{V}_m\right)\\ \boldsymbol{y}_m &= \mathbf{Q}_m\mathbf{U}_m/\mathbf{Q}_m\mathbf{G}_m.\end{aligned} \tag{11}$$

It can be seen that (10) and (11) are closely related. Specifically, $\boldsymbol{\xi}_i \sim \mathbf{U}_m$, $\mathbf{M}_i \sim \mathbf{K}_m$, $\mathbf{x}_i \sim \mathbf{V}_m$, and $\mathbf{N}_i \sim \mathbf{Q}_m$. As a result, selective SSM may be interpreted as a variant of

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linear attention, with a strong correlation between the two mechanisms.

Fig. 3(a) illustrates the architecture of the linear attention transformer, mainly composed of linear attention, normalization, and a multilayer perceptron (MLP). In the Mamba block, S6 is replaced by linear attention, and using an MLP, we obtain the MLLA. The MLLA schematic is visualized in Fig. 3(c). Specifically, we first designed a shortcut to enhance the stability of training, where the shortcut is similar to a residual block. Next, we replaced S6 with linear attention, used DSC instead of regular convolution, and incorporated the idea of residual learning to obtain the MLLA block. Finally, we used an MLP to enable the model to handle more complex data.

B. Embedding Module

The embedding module transforms the HSI from a high-dimensional representation into a low-dimensional representation, intending to effectively extract and represent important information in the image to improve classification performance. Unlike the existing patch-based embedding methods, this embedding layer extracts embeddings for each pixel in the image, allowing it to capture finer grained embeddings. This work also uses absolute positional encoding to enhance the positional dependencies between image pixels. Taking $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ as the HSI input, the embedding process of $\mathbf{X}$ is expressed using the subsequent formulas

$$\begin{aligned}\mathbf{E} &= \text{Embedding}(\mathbf{X}) \\ &= \text{SiLU}(\text{GN}(\text{Conv2d}(\mathbf{X}))) \\ \mathbf{A} &= \text{APE}(\mathbf{E}) \end{aligned} \tag{12}$$

where Conv2d is a 2-D convolution layer with a 1×1 kernel, GN is a group normalization layer, and SiLU is the activation function. H, W, and C represent the height, width, and spectral band of the input HSI, respectively. APE stands for absolute positional embedding. $\mathbf{E} \in \mathbb{R}^{H\times W\times P}$ represents the extracted embedding, and $\mathbf{A} \in \mathbb{R}^{H\times W\times P}$ represents the embedding values after absolute positional embedding. P represents the embedding dimension.

C. SpaMLLA Block

HSI processing requires consideration not only of the spectral information but also of the spatial structure, as the spectral features of adjacent pixels are often highly correlated. Since the HSIC is a pixel-by-pixel classification work, it requires the extraction of fine-grained features. Using the entire image as input, the model is better able to explore the complex relationships and structures within the image, leading to more accurate classification. In this study, the MLLA module is used as the basic building block. The SpaMLLA module is designed as the central part of the spatial branch, comprising three stacked SpaMLLA modules and pooling layers. This design allows for effective capture of the contextual dependencies between pixels while maintaining linear complexity. The specific structure is shown in Fig. 4. The forward process can be described using the following equation:

$$\mathbf{A}_{\mathbf{spa}} = \text{Proj}\left(\text{Reshape}\left(\text{MLLA}\left(\text{Reshape}\left(\mathbf{A}\right)\right)\right)\right) + \mathbf{A} \tag{13}$$

where $\mathbf{A} \in \mathbb{R}^{1\times P\times H\times W}$ represents the embedding features, and $\mathbf{A}_{\mathbf{spa}} \in \mathbb{R}^{1\times P\times H\times W}$ stands for the output features of SpaMLLA after introducing residual connections. And 1, P, H, and W represent the batch size of 1, the embedding dimension, the image height, and the image width, respectively. In the experiment, the embedding dimension is set to 128. MLLA refers to the MLLA module. Proj refers to projection, which is instrumental in enhancing the model's generalization ability.

D. SpeMLLA Block

The spectral branch classifies the image based on the spectral characteristics of each pixel. Each pixel in HSI usually includes reflectance information across a large number of spectral bands [53]. The spectral details enable the discrimination of various land cover classes. The spectral branch uses these spectral features to perform feature extraction, feature selection, and classification using various methods. Currently, feature extraction by modeling the relationships between spectral features remains a challenge. Unlike the spatial module, the spectral branch divides the spectral characteristics into G-groups and models the spectral relationships within each group. The spectral branch consists of three stacked SpeMLLA modules and pooling layers, as displayed in Fig. 4. The following formulas describe the forward propagation process:

$$\begin{aligned}\mathbf{G}_{\mathbf{spe}} &= \text{SplitGroup}(\mathbf{A}) \\ \mathbf{A}_{\mathbf{spe}} &= \text{Proj}\left(\text{Reshape}\left(\text{MLLA}\left(\text{Reshape}\left(\mathbf{G}_{\mathbf{spe}}\right)\right)\right)\right) + \mathbf{A} \end{aligned} \tag{14}$$

where $\mathbf{A} \in \mathbb{R}^{1\times P\times H\times W}$, $\mathbf{G}_{\mathbf{spe}} \in \mathbb{R}^{1\times G\times Q\times H\times W}$, and $\mathbf{A}_{\mathbf{spe}} \in \mathbb{R}^{G\times Q\times H\times W}$ represent the embedding dimension, the divided spectral group features, and the output features, respectively. Q denotes the embedding dimension P divided by the group count G, where G is fixed at 2 in the experiment.

E. SSF Module

Algorithm 1 Pseudoprocedure of DBMLLA

Input: Whole image $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ and embedding dimension P
Output: Class predictions on the test samples
Initialize: randomly initialize the model parameters;
`// Absolute positional embedding`
$\mathbf{E} \leftarrow$ Embedding($\mathbf{X}$);
$\mathbf{A} \leftarrow$ APE($\mathbf{E}$) in (12);
for $i = 1$ **to** n **do**
  `// Forward processing procedure`
  $\mathbf{A}_{\mathbf{spa}} \leftarrow$ SpaMLLA($\mathbf{A}$) in (13);
  $\mathbf{A}_{\mathbf{spe}} \leftarrow$ SpeMLLA($\mathbf{A}$) in (14);
  $\mathbf{A}_{\mathbf{fus}} \leftarrow$ Fusion($\mathbf{A}, \mathbf{A}_{\mathbf{spa}}, \mathbf{A}_{\mathbf{spe}}$) in (15);
end
Classifier:
Classify the feature map to generate predictions;

This module aims to improve classification accuracy by merging spatial and spectral features. In this article, the softmax function is applied to create fusion coefficients for

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TABLE I
COUNTRY OF COLLECTION, COLLECTION EQUIPMENT, SAMPLE SIZE, IMAGE SIZE, NUMBER OF CATEGORIES, AND NUMBER OF SPECTRAL BANDS OF THE FOUR DATASETS

Number	Dataset	Collection Location	Collection Equipment	Sample Size	Image Size	Categories	Spectral Bands
No.1	UP	Pavia, Italy	Reflective Optical System Imaging Spectrometer	42776	610 × 340	9	103
No.2	Hanchuan	Hubei Province, China	Headwall Nano-Hyperspec Imaging Sensor	257530	1217 × 303	16	274
No.3	Trento	Trento, Italy	AISA Eagle Sensor	30214	600 × 166	6	63
No.4	SV	Salinas Valley, California	AVIRIS Sensor	53063	512 × 217	16	204

TABLE II
NAMES AND QUANTITIES OF EACH CLASS IN THE UP DATASET

Class Name	Train	Validation	Test	Total
C1	63	63	6505	6631
C2	186	186	18277	18649
C3	20	20	2059	2099
C4	30	30	3004	3064
C5	13	13	1319	1345
C6	50	50	4929	5029
C7	13	13	1304	1330
C8	36	36	3610	3682
C9	9	9	929	947
Total	420	420	41936	42776

the spatial and spectral branches separately for fusion. The integration process is represented by the equation below

$$\mathbf{A_{fus}} = \mathbf{A} + w_{\text{spa}} \times \mathbf{A_{spa}} + w_{\text{spe}} \times \mathbf{A_{spe}} \tag{15}$$

where $\mathbf{A_{fus}}$ represents the fused output features. The fusion weights w_{spa} and w_{spe} are automatically learned through trainable parameters combined with the softmax function, representing the relative importance of the SpaMLLA and SpeMLLA modules in the final output. These weights are dynamically adjusted during training to enhance the model's performance. The spatial–spectral fusion (SSF) module fuses the important information from the spatial and spectral features to generate the definitive classification result. The procedural steps of the DBMLLA model are summarized in Algorithm 1.

IV. EXPERIMENTS

A. Datasets

Four HSI datasets—Pavia University (UP), WHU-Hi-Hanchuan (Hanchuan), Trento, and Salinas Valley (SV)—are introduced in this section. Table I shows the information on the four datasets, including the collection location, collection equipment, sample size, image size, categories, and spectral bands. The ratio of the training set is 1% of the available marked samples. The quantities of training samples, validation samples, and test samples are displayed in Tables II–V. Furthermore, Figs. 5–8 display the false-color images and ground truths on the four datasets, as well as the color for each category.

B. Experimental Settings

1) Evaluation Metrics: To evaluate the model's performance, this experiment applied three frequently used

TABLE III
NAMES AND QUANTITIES FOR EACH CLASS IN THE HANCHUAN DATASET

Class Name	Train	Validation	Test	Total
C1	447	447	43841	44735
C2	227	227	22299	22753
C3	102	102	10083	10287
C4	53	53	5247	5353
C5	12	12	1176	1200
C6	45	45	4443	4533
C7	59	59	5785	5903
C8	179	179	17620	17978
C9	94	94	9308	9496
C10	105	105	10316	10526
C11	169	169	16573	16911
C12	36	36	3607	3679
C13	91	91	8934	9116
C14	185	185	18190	18560
C15	11	11	1114	1136
C16	754	754	73893	75401
Total	2569	2569	252392	257530

TABLE IV
NAMES AND QUANTITIES FOR EACH CLASS IN THE TRENTO DATASET

Class Name	Train	Validation	Test	Total
C1	40	40	3954	4034
C2	29	29	2845	2903
C3	4	4	471	479
C4	91	91	9023	9123
C5	105	105	10291	10501
C6	31	31	29614	3174
Total	300	300	41936	30214

TABLE V
NAMES AND QUANTITIES FOR EACH CLASS IN THE SV DATASET

Class Name	Train	Validation	Test	Total
C1	19	19	1931	1969
C2	36	36	3580	3652
C3	19	19	1900	1938
C4	13	13	1342	1368
C5	26	26	2574	2626
C6	38	38	3805	3881
C7	35	35	3439	3509
C8	110	110	10827	11047
C9	60	60	5959	6079
C10	32	32	3150	3214
C11	10	10	1028	1048
C12	18	18	1853	1889
C13	8	8	882	898
C14	10	10	1030	1050
C15	71	71	6982	7124
C16	17	17	1737	1771
Total	522	522	52019	53063

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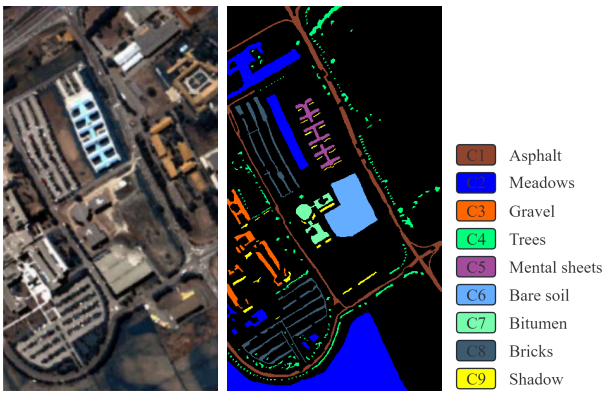


Fig. 5. UP dataset. (a) False-color map. (b) Ground truth. (c) Legends.

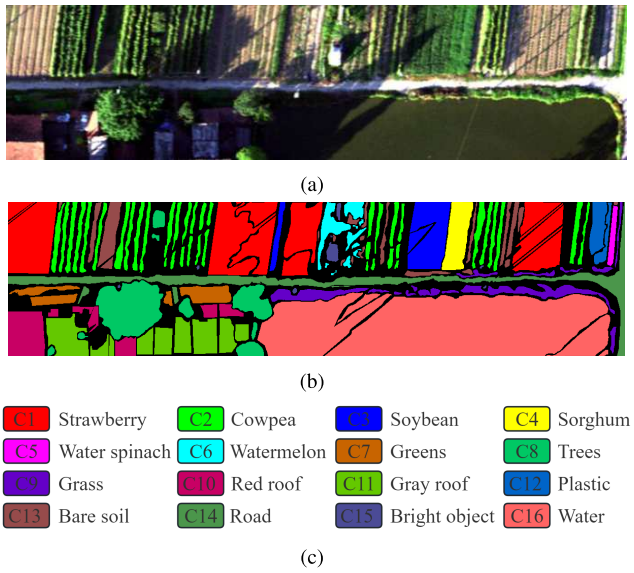


Fig. 6. Hanchuan dataset. (a) False-color map. (b) Ground truth. (c) Legends.



Fig. 7. Trento dataset. (a) False-color map. (b) Ground truth. (c) Legends.

performance metrics, including overall accuracy (OA), average accuracy (AA), and Kappa.

2) Comparison Methods: To validate the effectiveness of the proposed DBMLLA method, we conducted a comparative analysis against other advanced methods, including machine learning methods (SVM [11]), CNN-based methods (Tri-CNN [28]), transformer-based methods (SSFTT [34], MASSformer [54], GSC-VIT [55]), and Mamba-based methods (MambaHSI [42], 3DSS-Mamba [43], and SS-Mamba [45]).

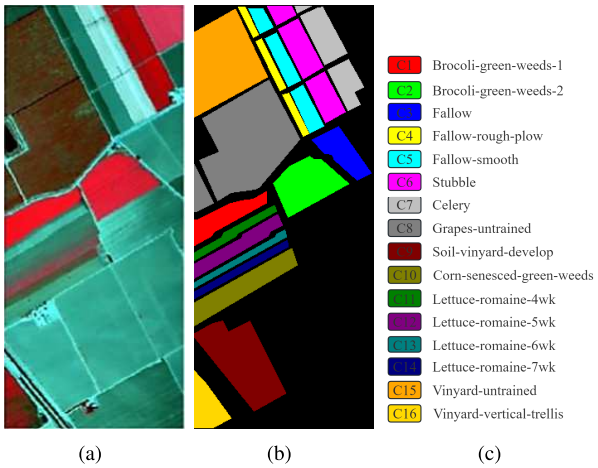


Fig. 8. SV dataset. (a) False-color map. (b) Ground truth. (c) Legends.

a) *SVM:* The SVM method is implemented using a Gaussian kernel on the libsvm platform, with SVM parameters optimized through fivefold cross-validation to solve the HSIC problem.
b) *Tri-CNN:* Tri-CNN effectively extracts multiscale and multimodal information through its multibranch structure and feature fusion.
c) *SSFTT:* The SSFTT method first uses 2-D and 3-D convolutional layers to extract shallow features, and then a Gaussian weighting method is introduced to extract deeper features.
d) *MASSformer:* The SSFTT method incorporates the memory tokenizer and the memory-augmented transformer encoder, constructing a memory-augmented transformer for HSIC.
e) *GSC-VIT:* The GSC-VIT method separately uses grouped pointwise convolution and ViT to capture local features and global information in HSI.
f) *MambaHSI:* The MambaHSI method uses Mamba to model the contextual dependencies of the image, marking the first time that Mamba is applied to HSIC.
g) *3DSS-Mamba:* The 3DSS-Mamba method introduces a 3-D selective scanning mechanism that can be used for HSIC in high-dimensional scenarios.
h) *SS-Mamba:* The SS-Mamba method first introduces a spectral–spatial learning framework based on the Mamba model. It achieves the extraction of features by designing a token generation mechanism and a feature enhancement module.

3) Implementation Details: The experiments have been conducted using the PyTorch platform. The hardware used is: NVIDIA GeForce RTX 4060 Ti GPU, x64 Intel[1] Core[2] i7-14700KF CPU, and 128-GB RAM. In the experiments, the

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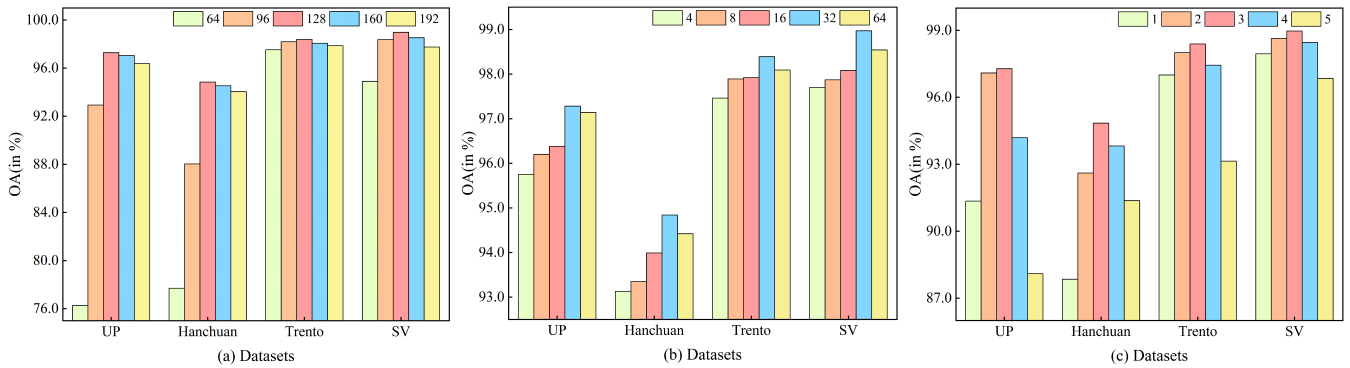


Fig. 9. Analysis of the overall classification performance obtained by our method with different parameters on four datasets. (a) Embedding dimension. (b) Number of attention heads. (c) Layer number in the MLLA module.

training and validation samples each accounted for 1% of all the samples, and the remaining samples are used as the test set. Since our proposed DBMLLA method takes the whole image as input, the batch size was set to 1. The quantity of epochs for each training session is set to 200. To minimize the impact of randomness and enhance the reliability and stability of the outcomes, all the experiments are based on the average results of ten consecutive trials.

C. Parameter Analysis

1) Analysis of the Embedding Dimension: Fig. 9(a) shows the classification accuracies on four datasets with varying embedding dimensions in the range of $\{64, 96, 128, 160, 192\}$. It is evident that increasing the embedding dimension initially enhances the model's learning capability and improves performance. Nevertheless, as the embedding dimension continues to increase, redundant features are introduced due to the limited number of training samples, which leads to overfitting and a decline in classification accuracy. Thus, the optimal embedding dimension selected in this work is 128.

2) Analysis of the Number of Attention Heads: The number of attention heads has a significant impact on the model's representation capacity and overall performance. In this work, we conduct experiments by varying the number of attention heads. Fig. 9(b) presents the classification results on four datasets, where the number of attention heads is set to $\{4, 8, 16, 32, 64\}$. It can be observed that the model performance first improves and then degrades as the number of heads increases, with the best result achieved when using 32 heads. This is because an insufficient number of attention heads may limit the model's ability to capture long-range dependencies. A larger number of heads allows the model to attend to more regions and extract richer contextual information. However, when the number of heads exceeds a certain threshold, additional heads may become redundant and fail to learn useful patterns, thus negatively impacting the classification performance. Therefore, the number of attention heads is set to 32 in our experiments.

3) Analysis of the Layer Number in the MLLA Module: To explore the optimal number of layers for the SpaMLLA and SpeMLLA modules in DBMLLA, we conducted a series of experiments with the number of layers set to $\{1, 2, 3, 4, 5\}$. Fig. 9(c) presents the OA on four datasets. The results show that the classification performance gradually improves as the number of layers increases, reaching the best performance when the number of layers is 3, which is significantly better than other settings. Increasing the number of layers reduces classification performance, as excessive depth leads to redundant or ineffective feature extraction, which interferes with the model's ability to learn discriminative features. Based on this observation, the number of layers in both the SpaMLLA and SpeMLLA modules is set to 3.

D. Comparison to SOTAs

The quantitative analyses [OA (%), AA (%), and Kappa (%)] of the UP, Hanchuan, Trento, and SV datasets are reported in Tables VI–IX and Figs. 10–13 display classification visualizations.

1) UP Dataset: For the UP dataset, the selected learning rate is 0.0003. Table VI provides an overview of the classification results for all the considered methods. With improvements of 1.31% in OA, 2.86% in AA, and 1.39% in Kappa, the proposed method outperforms the second-best SSFTT. Moreover, it achieves the highest classification accuracy in four categories among all the compared methods. As SVM disregards spatial information in HSI and Tri-CNN fails to effectively capture global features, their classification performance remains suboptimal. Transformer-based methods, such as SSFTT, establish long-range dependencies through attention mechanisms; however, their single-branch structure makes it difficult to fully capture both the spatial and spectral features simultaneously. Mamba is capable of mining discriminative spatial and spectral features, but its global modeling capability is inferior to that of the transformer. In comparison, DBMLLA exhibits stronger global modeling capabilities. SpaMLLA leverages linear attention to model global spatial contextual information, while SpeMLLA is designed to capture correlations between spectral bands. DBMLLA also uses a dual-branch structure to fully extract both the spatial and spectral information while minimizing information loss to the greatest extent.

Fig. 10 illustrates the classification maps obtained from all the methods used in this experiment. From the results, it is evident that methods based on SVM and CNN suffer from significant noise interference, while methods based on transformer and Mamba face issues related to edge information confusion. In contrast, our method produces smoother

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TABLE VI

CLASSIFICATION PERFORMANCE OF THE SVM, TRI-CNN, SSFTT, MASSFORMER, GSC-VIT, MAMBAHSI, 3DSS-MAMBA, SS-MAMBA, AND PROPOSED DBMLLA METHODS FOR THE UP DATASET

Class	Classification accuracy achieved by different models (in %)								
	SVM	Tri-CNN	SSFTT	MASSformer	GSC-VIT	MambaHSI	3DSS-Mamba	SS-Mamba	Proposed
C1	90.56	97.41	**98.60**	94.55	96.55	94.47	89.46	96.03	94.77
C2	97.25	98.65	**99.71**	99.15	98.98	96.44	96.77	93.11	98.09
C3	57.62	69.95	89.21	84.95	83.97	93.02	79.20	99.04	**99.31**
C4	80.31	88.03	86.65	85.02	94.21	89.96	81.87	**99.04**	92.58
C5	99.26	99.88	98.65	97.60	99.84	99.99	92.33	99.91	**99.99**
C6	67.06	88.48	97.87	96.58	96.32	98.62	78.17	99.25	**99.39**
C7	38.50	69.76	99.02	94.66	86.94	96.37	63.42	99.76	**99.88**
C8	81.67	83.50	93.19	91.39	93.56	94.44	83.05	**98.91**	97.95
C9	99.79	99.36	91.29	73.91	97.68	99.35	47.19	**99.97**	99.27
OA	86.45±1.41	92.94±1.12	95.97±0.89	94.81±0.88	95.37±0.90	95.77±0.90	88.06±1.76	95.14±2.66	**97.28±1.12**
AA	79.11±4.56	88.34±2.76	94.91±1.55	90.87±1.76	94.23±1.13	95.84±1.12	79.04±3.95	96.47±0.89	**97.77±0.69**
Kappa	81.66±2.01	90.56±1.44	96.11±10.18	93.37±1.18	95.18±1.20	95.00±2.24	84.09±2.35	94.98±3.39	**97.50±1.77**

TABLE VII

CLASSIFICATION PERFORMANCE OF THE SVM, TRI-CNN, SSFTT, MASSFORMER, GSC-VIT, MAMBAHSI, 3DSS-MAMBA, SS-MAMBA, AND PROPOSED DBMLLA METHODS FOR THE HANCHUAN DATASET

Class	Classification accuracy achieved by different models (in %)								
	SVM	Tri-CNN	SSFTT	MASSformer	GSC-VIT	MambaHSI	3DSS-Mamba	SS-Mamba	Proposed
C1	93.78	97.59	**98.33**	98.52	97.15	90.01	94.56	94.28	93.50
C2	78.16	91.93	91.64	91.34	92.59	82.20	85.86	86.16	**86.17**
C3	72.36	82.61	90.44	94.28	88.50	94.84	76.91	90.43	**86.17**
C4	91.73	97.16	97.40	96.83	94.82	98.46	91.89	**99.07**	97.72
C5	16.58	34.68	62.31	67.65	75.15	99.52	61.33	**100.00**	99.77
C6	21.33	47.36	62.18	66.07	49.06	77.02	46.36	87.81	**95.60**
C7	80.36	77.58	86.97	89.05	74.96	91.72	86.57	96.70	**97.94**
C8	68.24	89.82	89.60	91.68	76.55	72.21	79.09	83.94	**92.94**
C9	60.64	73.70	82.61	89.79	80.63	88.93	70.20	88.86	**93.70**
C10	89.43	96.83	96.30	96.38	94.94	94.16	94.51	95.86	**97.13**
C11	94.13	92.92	93.65	92.01	90.21	88.70	93.42	95.09	**96.56**
C12	21.52	25.85	74.89	73.05	52.75	88.82	76.50	97.78	**98.84**
C13	48.75	60.74	74.32	72.28	69.14	79.05	64.00	80.28	**81.36**
C14	80.74	93.16	93.37	93.14	85.87	89.47	88.78	89.50	**93.99**
C15	53.49	57.73	66.41	70.79	66.28	**94.31**	49.82	93.48	89.64
C16	98.52	98.59	99.33	99.71	99.30	98.28	**99.44**	96.41	98.83
OA	84.04±0.22	90.56±0.07	93.37±1.09	93.58±0.82	90.36±0.97	90.34±1.61	89.27±0.87	92.52±2.00	**94.84±0.81**
AA	66.80±0.91	76.14±0.82	84.99±3.29	86.41±2.07	80.49±2.45	89.23±1.41	78.70±2.27	92.24±0.87	**94.44±0.63**
Kappa	81.28±0.26	88.93±0.07	92.23±1.28	92.94±0.97	88.69±1.14	88.93±2.39	87.43±1.02	91.30±2.27	**93.95±0.95**

TABLE VIII

CLASSIFICATION PERFORMANCE OF THE SVM, TRI-CNN, SSFTT, MASSFORMER, GSC-VIT, MAMBAHSI, 3DSS-MAMBA, SS-MAMBA, AND PROPOSED DBMLLA METHODS FOR THE TRENTO DATASET

Class	Classification accuracy achieved by different models (in %)								
	SVM	Tri-CNN	SSFTT	MASSformer	GSC-VIT	MambaHSI	3DSS-Mamba	SS-Mamba	Proposed
C1	81.83	98.25	96.98	94.78	98.60	98.23	97.33	**99.98**	99.49
C2	83.55	86.76	91.98	91.33	82.54	91.25	80.45	88.89	**92.31**
C3	95.12	84.85	42.57	22.45	91.48	98.22	18.97	88.43	**98.60**
C4	96.90	99.84	99.94	99.98	99.33	99.94	99.51	99.99	**100.00**
C5	87.21	99.58	99.94	99.95	99.89	99.77	98.53	**100.00**	99.99
C6	61.58	89.50	90.31	90.81	88.62	87.09	79.57	84.36	**92.49**
OA	86.49±0.48	96.46±0.16	96.86±0.58	96.25±0.54	96.57±1.00	97.45±0.41	93.68±0.54	96.10±0.85	**98.39±0.53**
AA	84.36±0.97	93.13±1.20	86.95±2.33	83.22±1.66	93.41±2.28	95.75±0.57	79.06±2.35	93.61±1.88	**97.15±0.96**
KAPPA	82.04±0.65	95.94±0.21	95.80±0.78	94.98±0.73	95.41±1.33	96.20±2.27	91.53±0.73	96.13±1.14	**97.62±0.62**

classification maps with the least noise at the boundaries, demonstrating that our approach is the most competitive one.

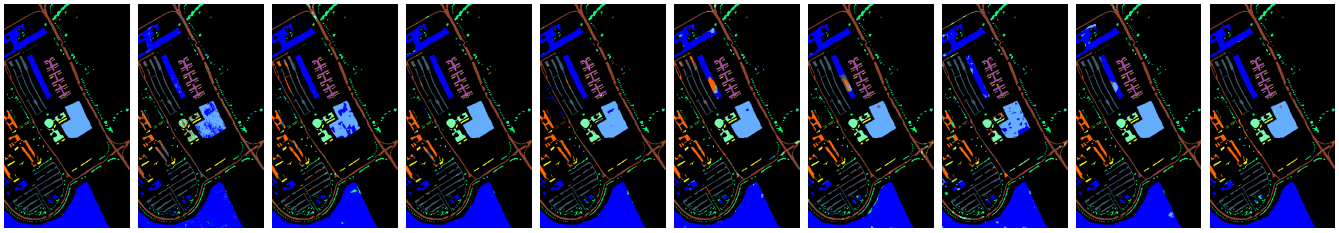
2) Hanchuan Dataset: With respect to the Hanchuan dataset, the learning rate is established at 0.0003. Table VII presents the classification accuracy of the nine comparative methods. It is evident from the classification results that the Mamba-based classification methods achieve lower accuracy than the transformer-based classification methods.

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TABLE IX

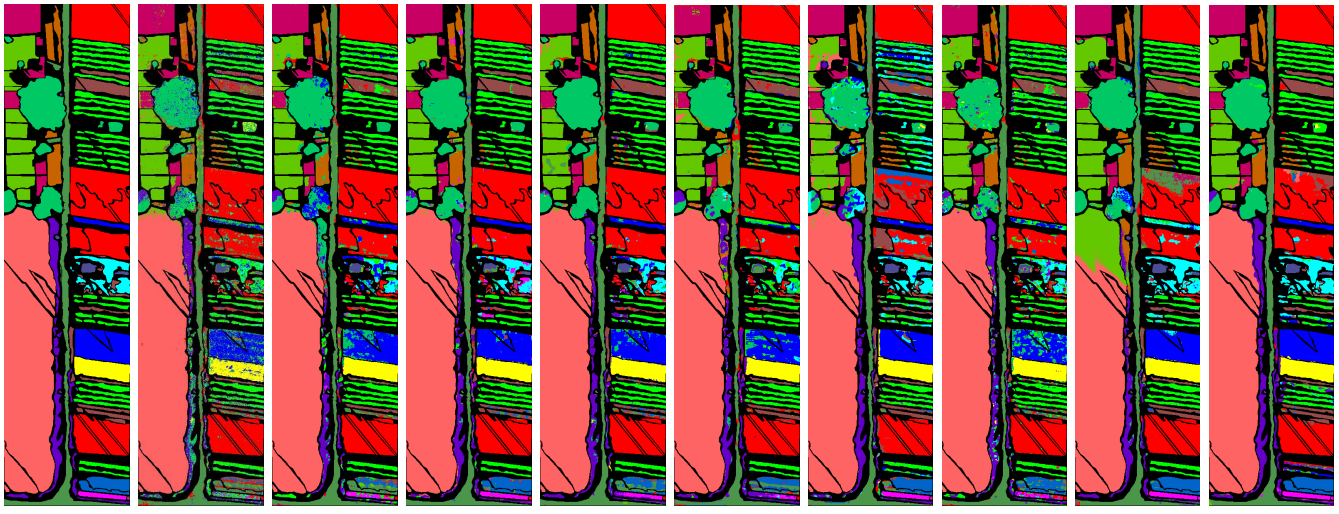
CLASSIFICATION PERFORMANCE OF THE SVM, TRI-CNN, SSFTT, MASSFORMER, GSC-VIT, MAMBAHSI, 3DSS-MAMBA, SS-MAMBA, AND PROPOSED DBMLLA METHODS FOR THE SV DATASET

Class	Classification accuracy achieved by different models (in %)								
	SVM	Tri-CNN	SSFTT	MASSformer	GSC-VIT	MambaHSI	3DSS-Mamba	SS-Mamba	Proposed
C1	94.85	99.80	99.73	99.99	96.59	99.97	97.87	88.90	**100.00**
C2	98.91	99.98	99.99	99.90	99.61	99.99	98.98	93.81	**100.00**
C3	95.02	99.62	99.43	97.63	88.77	**99.93**	94.66	99.55	99.68
C4	99.15	98.01	95.58	91.62	97.88	**99.90**	97.26	99.67	99.82
C5	96.72	96.43	95.24	97.89	95.05	98.26	97.16	98.10	**98.34**
C6	99.10	99.95	97.22	97.51	99.80	99.98	99.23	99.95	**99.98**
C7	99.46	99.98	98.61	99.02	99.82	99.78	95.03	**99.97**	99.88
C8	88.70	94.97	**98.49**	97.66	88.74	95.80	91.83	92.48	96.84
C9	98.76	99.89	99.93	99.63	99.20	99.97	99.64	99.99	**100.00**
C10	91.13	**99.51**	99.04	98.37	93.17	97.55	93.53	98.98	98.22
C11	94.14	99.68	83.68	81.02	81.78	99.92	90.82	98.06	**99.96**
C12	99.82	96.89	95.28	96.49	98.64	99.99	89.31	99.72	**99.99**
C13	97.66	99.03	81.90	83.11	97.39	99.82	66.16	97.09	**99.94**
C14	92.68	94.01	98.47	88.64	94.46	**99.91**	86.87	97.90	99.76
C15	57.67	80.32	91.97	97.53	82.96	97.73	87.21	84.03	**99.03**
C16	97.26	99.31	95.34	97.36	93.00	99.91	89.68	98.15	**99.92**
OA	90.15±0.69	95.75±0.71	96.88±2.31	97.28±1.29	93.18±1.16	97.54±0.53	93.47±0.90	95.06±3.27	**98.97±0.44**
AA	93.82±0.66	97.34±0.48	95.62±1.97	95.21±2.26	94.18±0.84	99.28±0.22	92.20±1.49	96.65±2.79	**99.45±0.21**
Kappa	89.01±0.78	95.26±0.80	96.52±2.59	96.97±1.44	92.40±1.29	99.00±1.16	92.72±1.00	94.50±3.65	**99.60±0.53**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 10. Classification visualizations for the UP dataset based on 1% training data per category. (a) Ground truth. (b) SVM, OA = 86.45%. (c) Tri-CNN, OA = 92.94%. (d) SSFTT, OA = 95.97%. (e) MASSformer, OA = 94.81%. (f) GSC-VIT, OA = 95.37%. (g) MambaHSI, OA = 95.77%. (h) 3DSS-Mamba, OA = 88.06%. (i) SS-Mamba, OA = 95.14%. (j) Proposed, OA = 97.28%.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 11. Classification visualizations for the Hanchuan dataset based on 1% training data per category. (a) Ground truth. (b) SVM, OA = 84.04%. (c) Tri-CNN, OA = 90.56 %. (d) SSFTT, OA = 93.37%. (e) MASSformer, OA = 93.58%. (f) GSC-VIT, OA = 90.36%. (g) MambaHSI, OA = 90.34 %. (h) 3DSS-Mamba, OA = 89.27%. (i) SS-Mamba, OA = 92.52%. (j) Proposed, OA = 94.84%.

This can be attributed to Mamba's limited capacity for global modeling compared with the transformer. Compared with the second-best MASSformer, the proposed method improves the three metrics by 1.26%, 8.03%, and 1.01%, respectively. Notably, the DBMLLA method achieves the highest accuracy in up to 11 classes among all the

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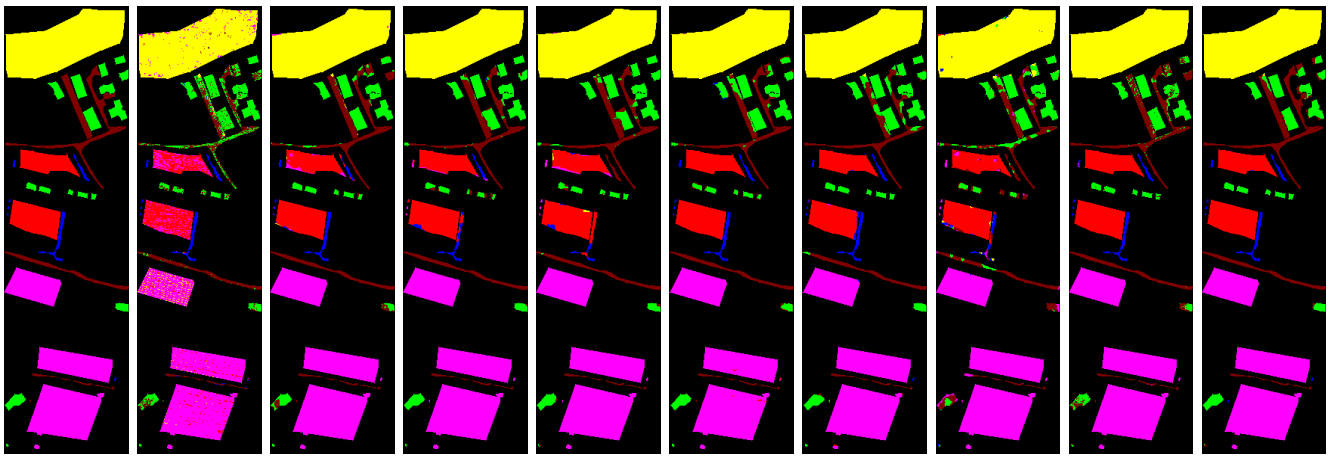


Fig. 12. Classification visualizations for the Trento dataset based on 1% training data per category. (a) Ground truth. (b) SVM, OA = 86.49%. (c) Tri-CNN, OA = 96.96%. (d) SSFTT, OA = 96.86%. (e) MASSformer, OA = 96.25%. (f) GSC-VIT, OA = 96.57%. (g) MambaHSI, OA = 97.45%. (h) 3DSS-Mamba, OA = 93.68%. (i) SS-Mamba, OA = 96.10%. (j) Proposed, OA = 98.39%.

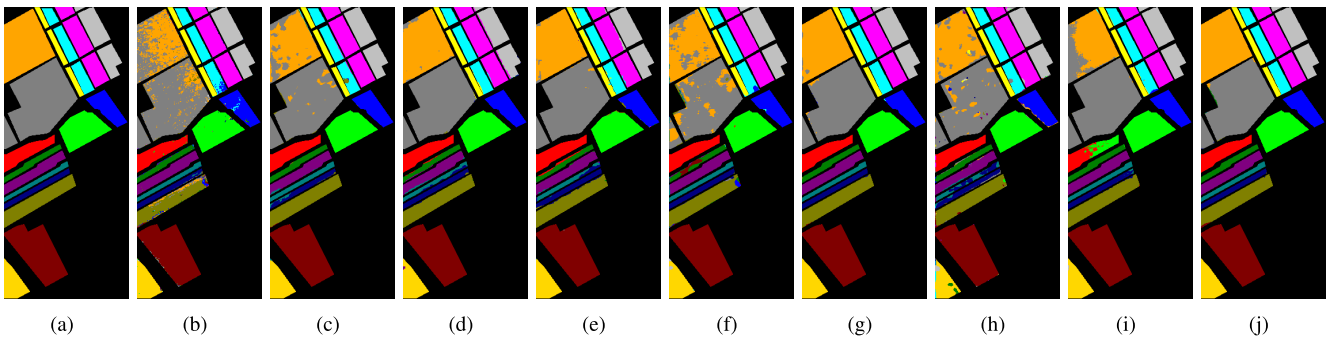


Fig. 13. Classification visualizations for the SV dataset based on 1% training data per category. (a) Ground truth. (b) SVM, OA = 90.15%. (c) Tri-CNN, OA = 95.75%. (d) SSFTT, OA = 96.88%. (e) MASSformer, OA = 97.28%. (f) GSC-VIT, OA = 93.18%. (g) MambaHSI, OA = 97.54%. (h) 3DSS-Mamba, OA = 93.47%. (i) SS-Mamba, OA = 95.06%. (j) Proposed, OA = 98.97%.

methods. This further reveals the effectiveness of the MLLA module in global modeling. The classification visualizations of all the modules are presented in Fig. 11. Benefiting from the comprehensive extraction of spectral–spatial semantic information, the proposed method has very few noise points, and the edge information confusion is also relatively minimal.

3) Trento Dataset: In relation to the Trento dataset, the learning rate is fixed at 0.0025. Table VIII presents the classification accuracy outcomes for all the research methods. Limited by modeling capabilities, the classification performance of 3DSS-Mamba is constrained. The latest SS-Mamba approach, which divides HSI samples into tokens, destroys the semantic structure of the samples, resulting in missing contextual information. Compared with DBMLLA, SS-Mamba decreases by 2.29%, 3.54%, and 1.49% in OA, AA, and Kappa. Among all the methods, the proposed DBMLLA approach achieved the best accuracy in four classes. For the classification result maps in Fig. 12, SVM suffers from more noise because it ignores spatial features, and other comparative methods exhibit confusion at the boundaries. The proposed method shows only slight confusion at the boundaries between the Ground class and the Roads class, while the rest of the areas remain very smooth.

4) SV Dataset: In the case of the SV dataset, the learning rate parameter is assigned a value of 0.00065. Table IX displays the classification accuracy of the nine methods used for comparison. In comparison to the second-best MambaHSI, the proposed method increases OA, AA, and Kappa by 1.43%, 0.17%, and 0.60%, respectively. Within these, the accuracy of ten categories is the highest of all the methods, and the accuracy of three categories even reaches 100%. Fig. 13 further provides the visualization maps of all the methods. As can be seen, SVM, Tri-CNN, GSC-VIT, and 3DSS-Mamba all have very significant noise, and the proposed method demonstrates greater smoothness. These phenomena explain the superiority of the proposed DBMLLA method.

E. Ablation Study

1) Ablation Study on DSC: DSC not only reduces the computational cost but also enhances the deep network performance. In the proposed framework, traditional convolutions are replaced with the more efficient DSC. To demonstrate the effectiveness of DSC, we keep other parameters unchanged and compare the classification results using DSC and standard convolution. Table X displays the OA for the four HSI datasets. From the results, it is evident that the application of DSC leads to improved classification performance, as DSC allows for the

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TABLE X

ABLATION STUDY OF THE KEY COMPONENTS OF OUR METHOD. DSC AND APE REPRESENT DSC AND ABSOLUTE POSITION ENCODING, RESPECTIVELY

DSC	APE	The overall accuracy of different datasets			
		UP	Hanchuan	Trento	SV
×	×	95.55	91.46	96.93	97.69
√	×	96.37	93.31	97.72	98.26
×	√	96.24	93.66	97.41	98.11
√	√	**97.28**	**94.84**	**98.39**	**98.97**

TABLE XI

ABLATION STUDY ON THE MAIN COMPONENTS OF OUR METHOD: SPAMLLA, SPEMLLA, AND FUSION (SIMPLE FUSION)

Spa	Spe	Fusion	SSF	UP	Hanchuan	Trento	SV
√	×	×	×	95.69	93.35	97.42	98.36
×	√	×	×	88.93	85.00	96.65	96.28
√	√	√	×	96.15	93.72	97.98	98.64
√	√	×	√	**97.28**	**94.84**	**98.39**	**98.97**

construction of deeper networks and the extraction of deeper features with the same computational resources.

2) Ablation Study on Absolute Position Encoding: The spatial relationships in HSI are crucial for HSIC. When handling sequential data, absolute position encoding enhances the model's capacity to understand the relationships between sequence elements, allowing the model to consider their relative positions, thereby improving its understanding of complex patterns. To enable the model to more effectively learn the spatial structure of HSI pixels, we combine absolute positional encoding with positional embeddings. To evaluate the impact of absolute positional encoding, we held all other parameters constant and analyzed the differences in classification outcomes between models with and without this feature. Table X provides the results for the four datasets used in this study. The experimental results confirm that the implementation of absolute positional encoding led to measurable enhancements in classification outcomes throughout all the four tested datasets. In summary, absolute positional encoding effectively enhances the model's expressive power.

3) Ablation Study on Fusion Module: Within this segment, we analyze the impact of SpaMLLA, SpeMLLA, simple fusion, and the proposed SSF module on classification performance. Table XI presents the classification results of the four configurations across four datasets. As shown in the table, the spatial branch outperforms the spectral branch by a large margin, indicating that spatial features are more discriminative than spectral features. After applying simple fusion, the classification performance improves. Furthermore, the proposed SSF module brings additional improvements, with classification accuracy increasing to varying degrees across all the datasets. This is because the proposed fusion strategy can adaptively integrate the most representative spatial and spectral features. This further demonstrates that spatial and spectral features are complementary, and their effective fusion leads to optimal performance.

4) Ablation Study on MLLA Module: The MLLA module is designed to enhance the model's ability to capture global contextual dependencies. To verify the effectiveness of the proposed MLLA module, we conducted comparative experiments between the MLLA and Mamba modules. Table XII presents the classification accuracies of the two models. As shown, compared with the Mamba module, the MLLA module achieves higher classification accuracy on the four datasets by 1.53%, 4.50%, 0.94%, and 0.53%, respectively. This improvement is attributed to the stronger global modeling capability of the MLLA module, further demonstrating the superiority of the proposed model.

TABLE XII

ABLATION STUDY OF THE MAMBA AND MLLA MODULES

Model	UP	Hanchuan	Trento	SV
Mamba	95.75	90.34	97.45	98.44
MLLA	**97.28**	**94.84**	**98.39**	**98.97**

F. Robustness Assessment

To evaluate the robustness of the proposed DBMLLA method, extensive experiments are conducted under varying proportions of training samples. The results of all the comparison methods on four datasets are presented in Fig. 14, where the performance of DBMLLA is indicated by the red line. The proportion of training samples was set between 0.1% and 2% for all the datasets. As shown in the results, the classification accuracy of DBMLLA consistently improves as the training sample ratio increases, demonstrating strong robustness. Benefiting from the powerful global modeling capability of the MLLA module, the proposed method outperforms other comparison methods at most sampling ratios. This further validates the superiority of the proposed approach.

G. Computational Complexity Analysis

We compared the number of parameters, training time, and testing time of all the methods on the UP dataset. The number of parameters refers to the total number of learnable parameters in each network. Table XIII presents the experimental results of all the methods. Although SVM and CNN-based methods exhibit fast training speeds, their performance is inferior due to the neglect of spatial information and the limitations of local receptive fields. Transformer-based architectures suffer from long testing times due to the quadratic computational complexity of the attention mechanism (for example, the testing time of GSC-VIT reaches 16.04 s). Mamba-based methods benefit from linear complexity and train relatively faster, but their performance is slightly lower. Our proposed method and MambaHSI take the entire HSI data as input, which avoids redundant computation and repeated model calls, resulting in faster testing time. Moreover, our testing time is faster than that of MambaHSI. Although its complexity is slightly higher than other methods, our approach consistently achieves significantly better classification accuracy across all the datasets. The additional computational cost is reasonable and acceptable. In conclusion, the proposed DBMLLA model strikes a balance between computational efficiency and classification accuracy.

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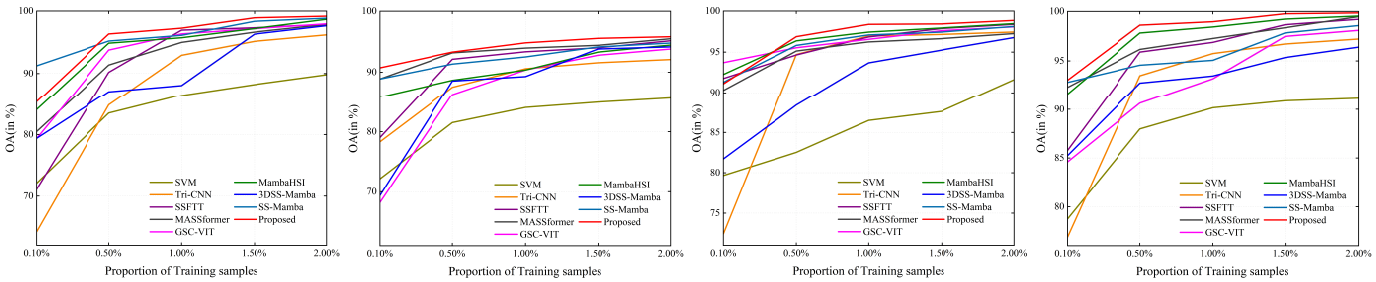


Fig. 14. Classification performance comparison under different proportions of labeled data on four datasets. (a) UP. (b) Hanchuan. (c) Trento. (d) SV.

TABLE XIII
COMPLEXITY ANALYSIS OF VARIOUS COMPARISON METHODS IN TERMS OF PARAMETER COUNT, TRAINING TIME, AND TESTING TIME

Metrics	SVM	Tri-CNN	SSFTT	MASSformer	GSC-VIT	MambaHSI	3DSS-Mamba	SS-Mamba	Proposed
Params (M)	−−	0.718	0.153	0.319	0.173	0.412	0.024	0.471	0.675
Training time (s)	0.26	22.05	5.98	12.59	29.91	418.04	42.92	14.66	259.71
Testing time (s)	1.36	13.41	1.19	2.17	16.04	0.42	3.95	0.02	0.24

V. CONCLUSION

This article puts forward a new DBMLLA for HSIC. Benefiting from the global modeling capability and linear complexity of our newly proposed MLLA, the proposed DBMLLA models global contextual information for HSI while maintaining linear computational complexity. In the SS-MLLA module, SpaMLLA is designed as the basic module for the spatial branch to extract spatial features, while SpeMLLA extracts spectral features by capturing correlations between different spectral groups. Simultaneously, the absolute position encoding captures spatial correlations between pixels, while DSC helps extract deeper features, and the fusion module enables efficient integration of spatial and spectral branches. Numerous experiments confirm that the proposed DBMLLA method exhibits stronger global modeling capabilities than advanced approaches.

The DBMLLA method offers added value for HSI tasks, including hazard risk assessment, medical imaging, and diagnosis. However, it has the limitation of being dependent on complex parameter settings. Future research will primarily focus on automatically optimizing these parameters and exploring more efficient HSIC frameworks.

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